BIOMEDICAL DATA DESIGN I

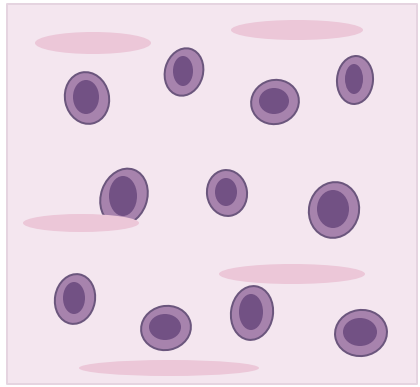
Nuclei segmentation project update

PM-Net, center detection, and what comes next

Zhiyu Song, Xinhao Jin, Yuchen Ai, Zheng Jian

THE IDEA

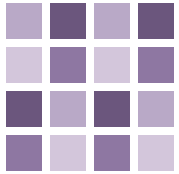
Architecture



01

H&E image

input tissue patch



UNI

image patterns



CNN adapter

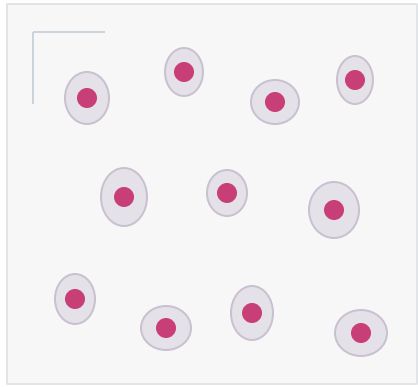
fine details

proposed

02

Image features

UNI + proposed detail branch

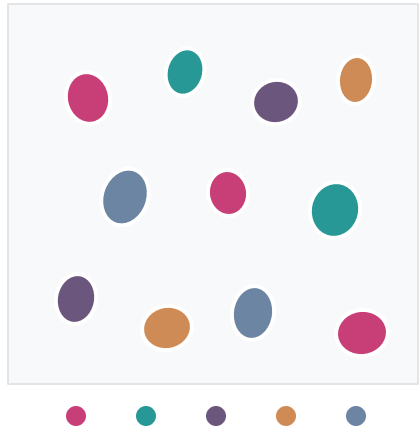


03

Find centers

one point per candidate nucleus

CURRENT FOCUS



04

Masks + types

one region and class per nucleus

Center-point focal loss

$$L = \Sigma_i \ell_i \; / \; \max(1, N)$$

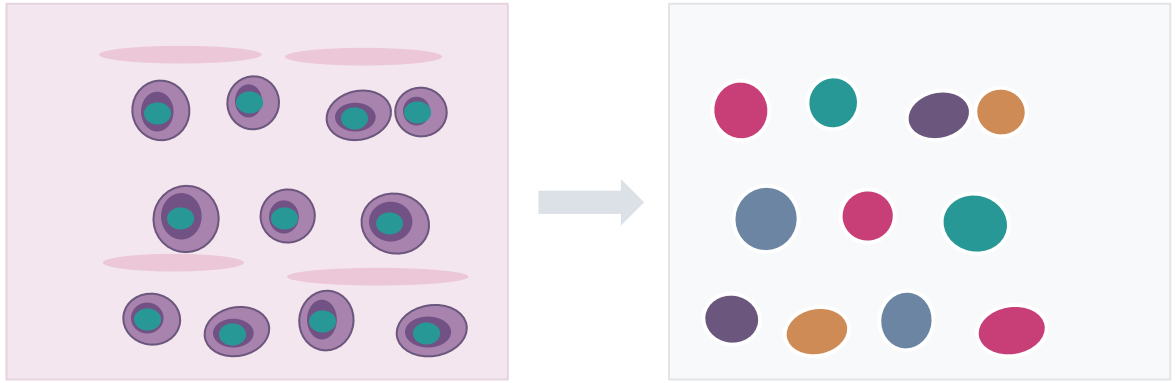
$$\ell_i = (1 - p_i)^2 \, [-\log(p_i)] \qquad \text{if } y_i = 1$$

$$\ell_i = (1 - y_i)^4 \, p_i^2 \, [-\log(1 - p_i)] \qquad \text{if } y_i < 1$$

p_i: predicted center probability y_i: target heatmap value N: nuclei in the batch

Where did the model struggle

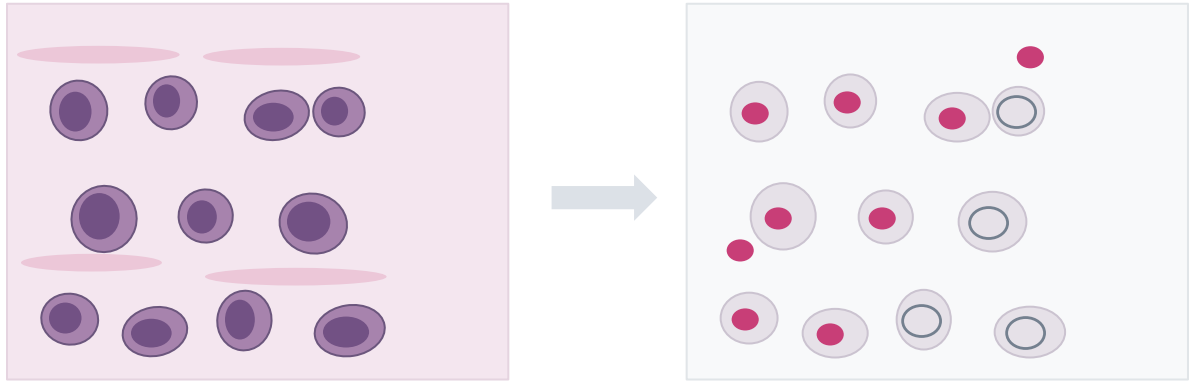
REAL LOCATIONS SUPPLIED



The model could learn many masks

Locations were provided in the diagnostic test

MODEL FINDS LOCATIONS



It missed nuclei or added wrong candidates

The center-finding step needed a separate test

CENTER TEST

Can the model find each nucleus center?

All nucleus classes are included. Dead is checked as a subgroup later.

1	**Teach**	Train on 2,134 labeled images for 130 epochs
2	**Predict**	Place a point near each nucleus
3	**Check**	Use 522 other images. Within 3 pixels counts as found.

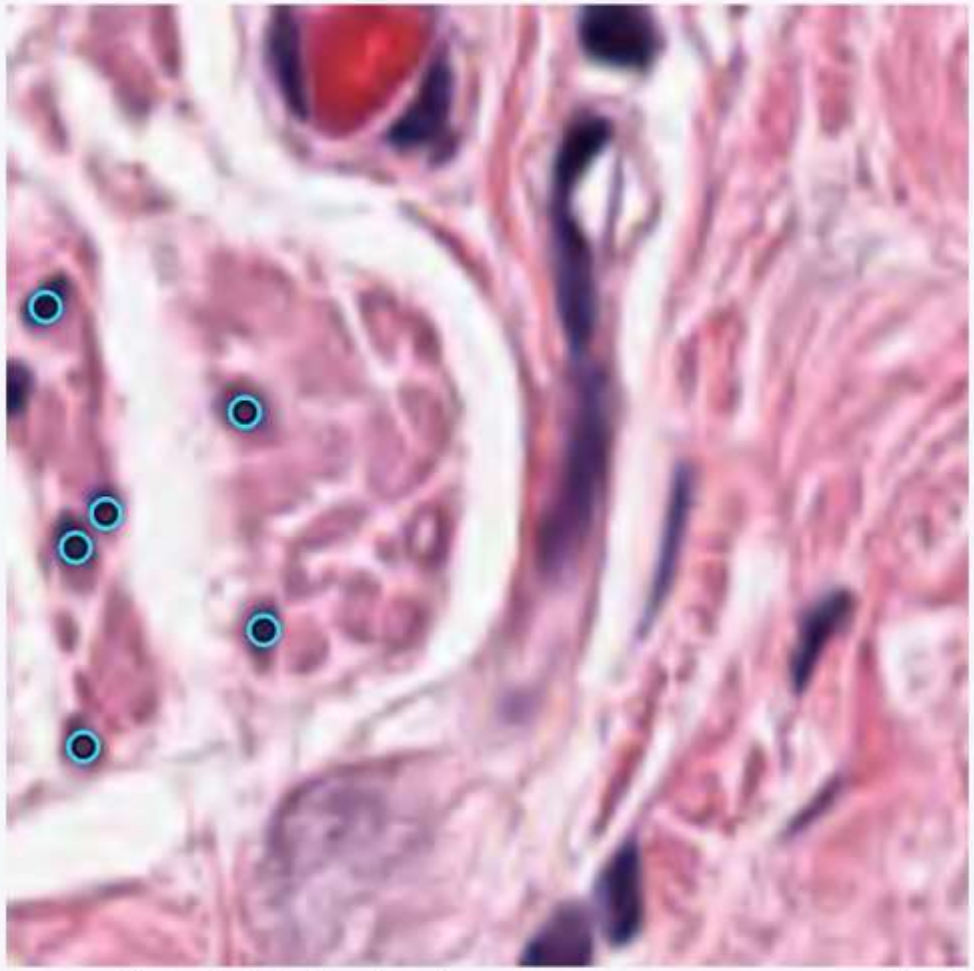
Recall: how many real nuclei were found.

Precision: how many predicted points were right.

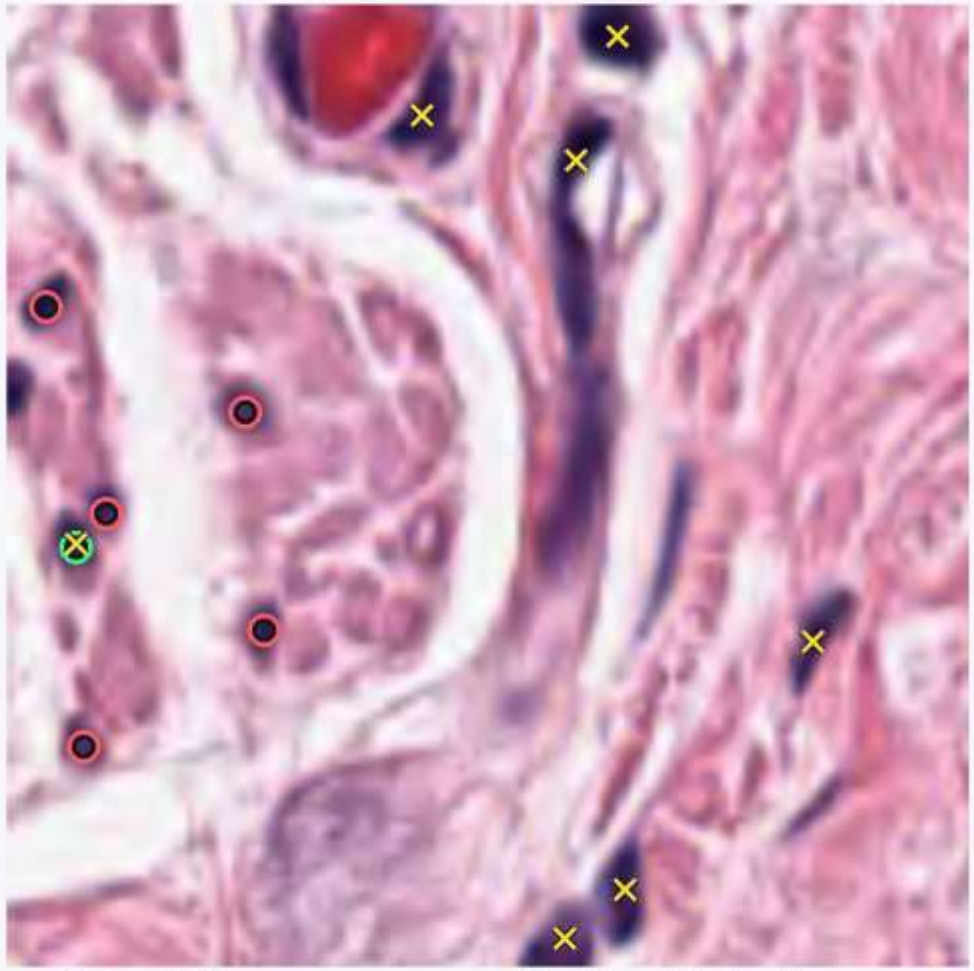
POINT DETECTION / UNI2-h P1R1

Image 155: 1 of 6 interior Dead nuclei detected

GROUND TRUTH



GROUND TRUTH + PREDICTIONS



cyan: labeled interior Dead nucleus green: matched GT red: missed GT yellow X: predicted point

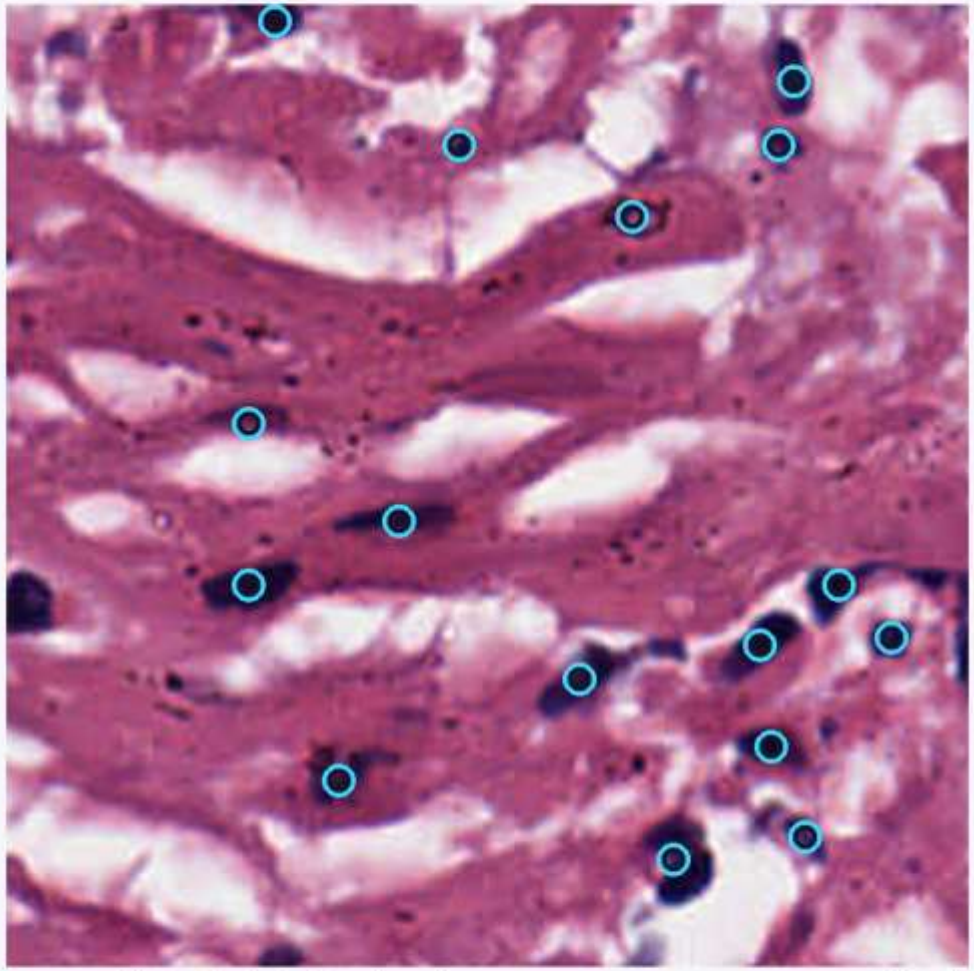
A GT point is matched if any predicted point is within 3 pixels. Predictions are not type-specific.

PanNuke fold 2 validation | P1R1 UNI2-h, seed 19, epoch 55 | selected examples, not overall performance

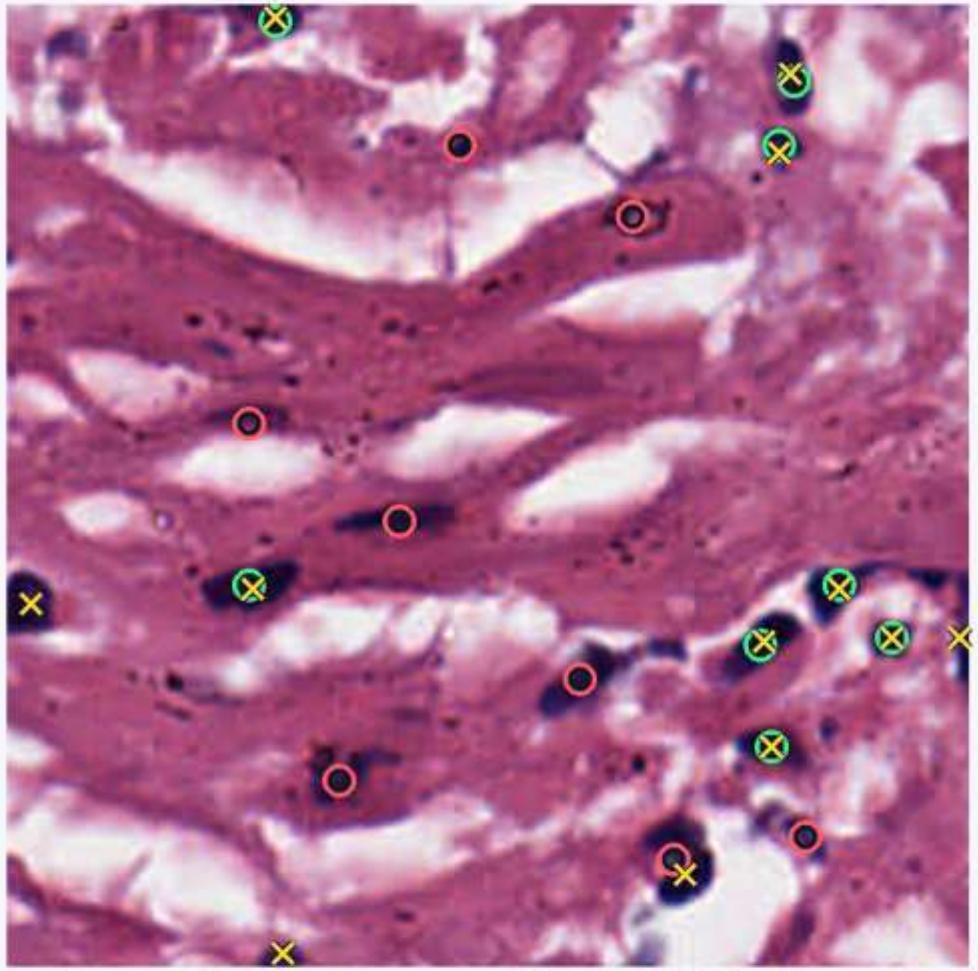
POINT DETECTION / UNI2-h P1R1

Image 1817: 8 of 16 interior Dead nuclei detected

GROUND TRUTH



GROUND TRUTH + PREDICTIONS



cyan: labeled interior Dead nucleus green: matched GT red: missed GT yellow X: predicted point

A GT point is matched if any predicted point is within 3 pixels. Predictions are not type-specific.

PanNuke fold 2 validation | P1R1 UNI2-h, seed 19, epoch 55 | selected examples, not overall performance

CENTER TEST RESULT

Good on fit images, much weaker on test images

98%

Nuclei found on fit images

59%

Nuclei found on test images

On test images: about 69% of predicted points were correct.

Dead subgroup: about 44% of centers were found.

FULL MODEL TEST

The tested full model scored far below baseline

mPQ measures the quality of the final masks and their class labels.



WHAT WE TRIED

Several fixes, but no stable result yet

More teaching	Foreground labels, Dead-center weight, wider positive region
Image and detail	Finer grid, broader center target, color changes
Training changes	Loss balance, fewer updated UNI layers, dropout

NEXT STEPS

Find out why center predictions fail on new images

01 **Sort the point errors**

Missed nuclei, duplicate points, misplaced points

02 **Check new ideas on more independent images**

The current internal audit set has been reused many times

03 **Then return to masks**

Connect a reliable detector and measure final segmentation